Final report submitted to Samson International PLC, Akuressa Road, Bogahagoda, Angulana, Galle.

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Smart Production Hourly Counting and History Management with Real-Time Monitoring on Things-Board Dashboard:

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1. Project Overview

This project aims to develop a low-cost, real-time production output monitoring system using the timer integrated with the ThingsBoard IoT platform. The system will allow foremen to set up the daily procedures as user inputs by logging into their ThingsBoard accounts via the mobile app (ThingsBoard Live). Supervisors and staff can review the real-time production (total count, day count, night count), hourly count as a bar graph, today's assignee/s (D/N), operation moulds (D/N), specific items being produced, color, target, and more. The foremen can also set the curing time via the dashboard. Since this operates via Wi-Fi, if any maintenance occurs, there is an option to correct the daily count (day count and night count) to maintain accuracy and reliability in case of any mismatches. However, when making corrections for both day and night, a specific table will be updated automatically to prevent any prohibited actions by either the operators or the foremen. In addition, the dashboard history widget maintains daily records as a time series for up to one month. This project was initially developed for the **E-25 machine** in the moulding section, which is used to make bathmats, and other items.

2. Requirements Fulfilled

- Developed a simple and reliable monitoring system.
- Transmit and receive both user input and production data to the Things-Board platform using MQTT protocol.
- Visualize hourly count in a bar graph to get the production status.
- Enable historical data logging for performance analysis and reporting.

3. System Components and Technologies

- Hardware: MCU
  - - Microcontroller (e.g., ESP32)
- Software:
  - - VS Code, Platform IO (For programming the microcontroller ESP32)
  - - Things-Board (Demo)
  - - MQTT protocol for data transfer
  - - Continuous Wi-Fi connection

4. System Architecture

4.1 Circuit Design on Easy EDA

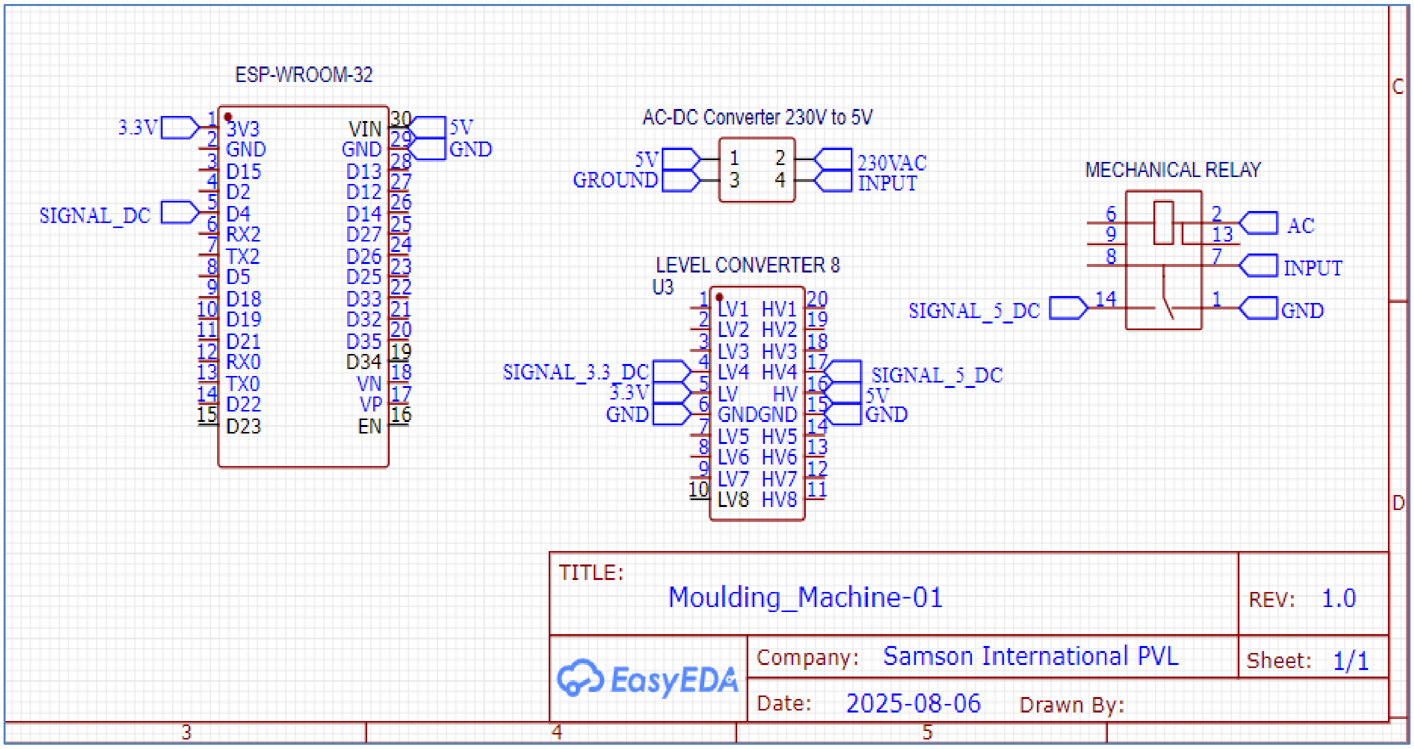


Figure 1: EASY EDA Circuit Diagram

4.2 Physical implemented Circuit

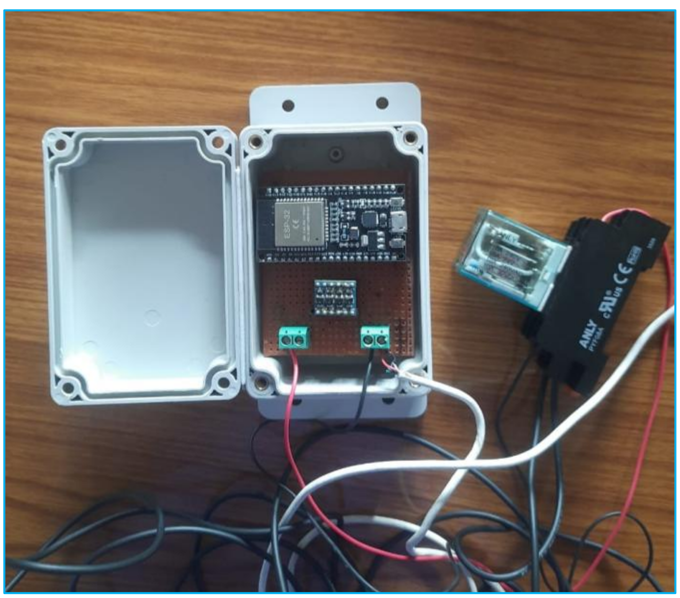
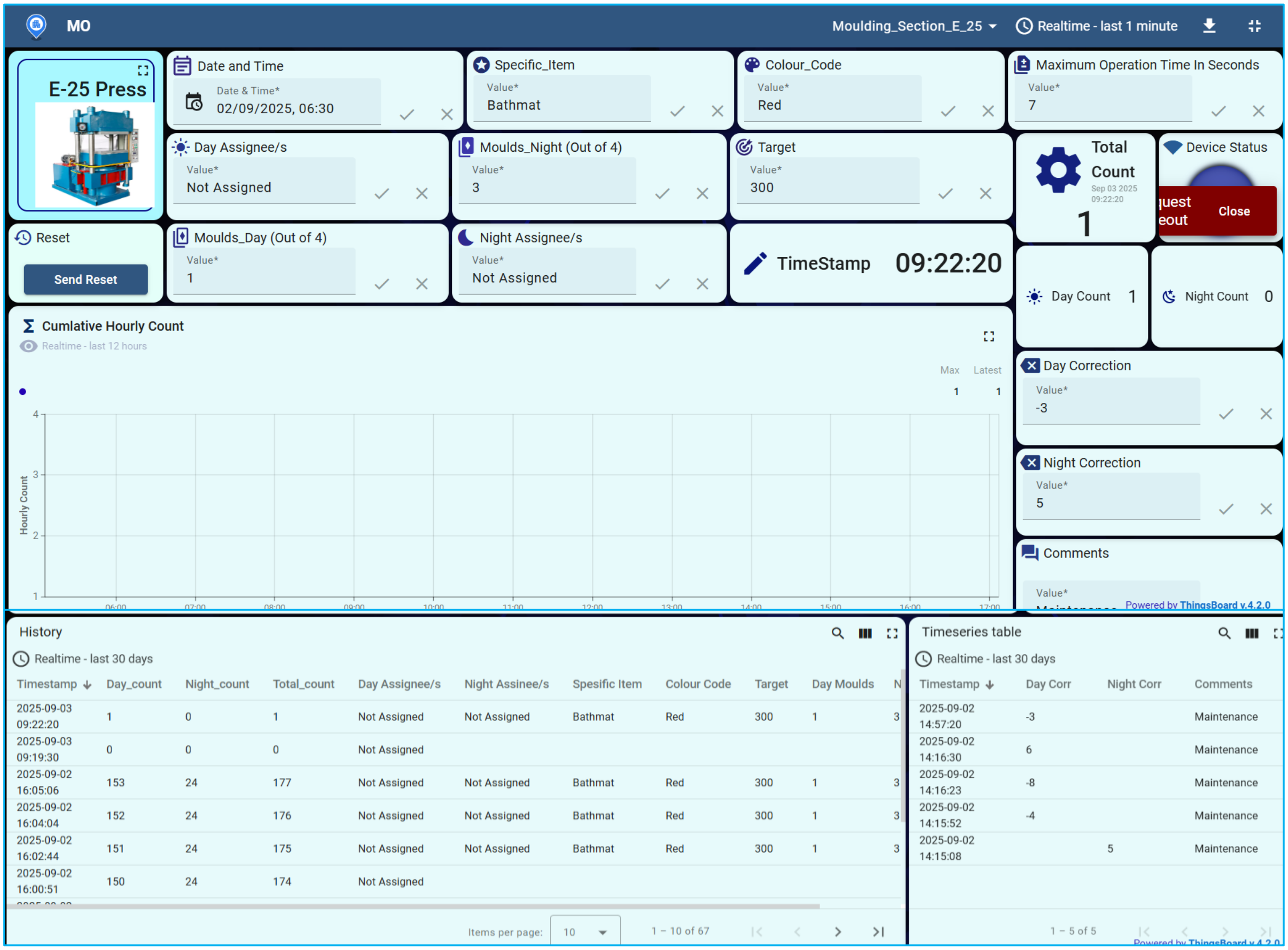


Figure 2: Dot board circuit

4.3 Things-Board Dashboard – Web App



Timestamp	Day_count	Night_count	Total_count	Day Assignee/s	Night Assinee/s	Spesific Item	Colour Code	Target	Day Moulds	N
2025-09-03 09:22:20	1	0	1	Not Assigned	Not Assigned	Bathmat	Red	300	1	3
2025-09-03 09:19:30	0	0	0	Not Assigned						
2025-09-02 16:05:06	153	24	177	Not Assigned	Not Assigned	Bathmat	Red	300	1	3
2025-09-02 16:04:04	152	24	176	Not Assigned	Not Assigned	Bathmat	Red	300	1	3
2025-09-02 16:02:44	151	24	175	Not Assigned	Not Assigned	Bathmat	Red	300	1	3
2025-09-02 16:00:51	150	24	174	Not Assigned						

Timestamp	Day Corr	Night Corr	Comments
2025-09-02 14:57:20	-3		Maintenance
2025-09-02 14:16:30	6		Maintenance
2025-09-02 14:16:23	-8		Maintenance
2025-09-02 14:15:52	-4		Maintenance
2025-09-02 14:15:08		5	Maintenance

Figure 3: Things-Board Dashboard – Web App

4.4 Things-Board Dashboard Mobile App.

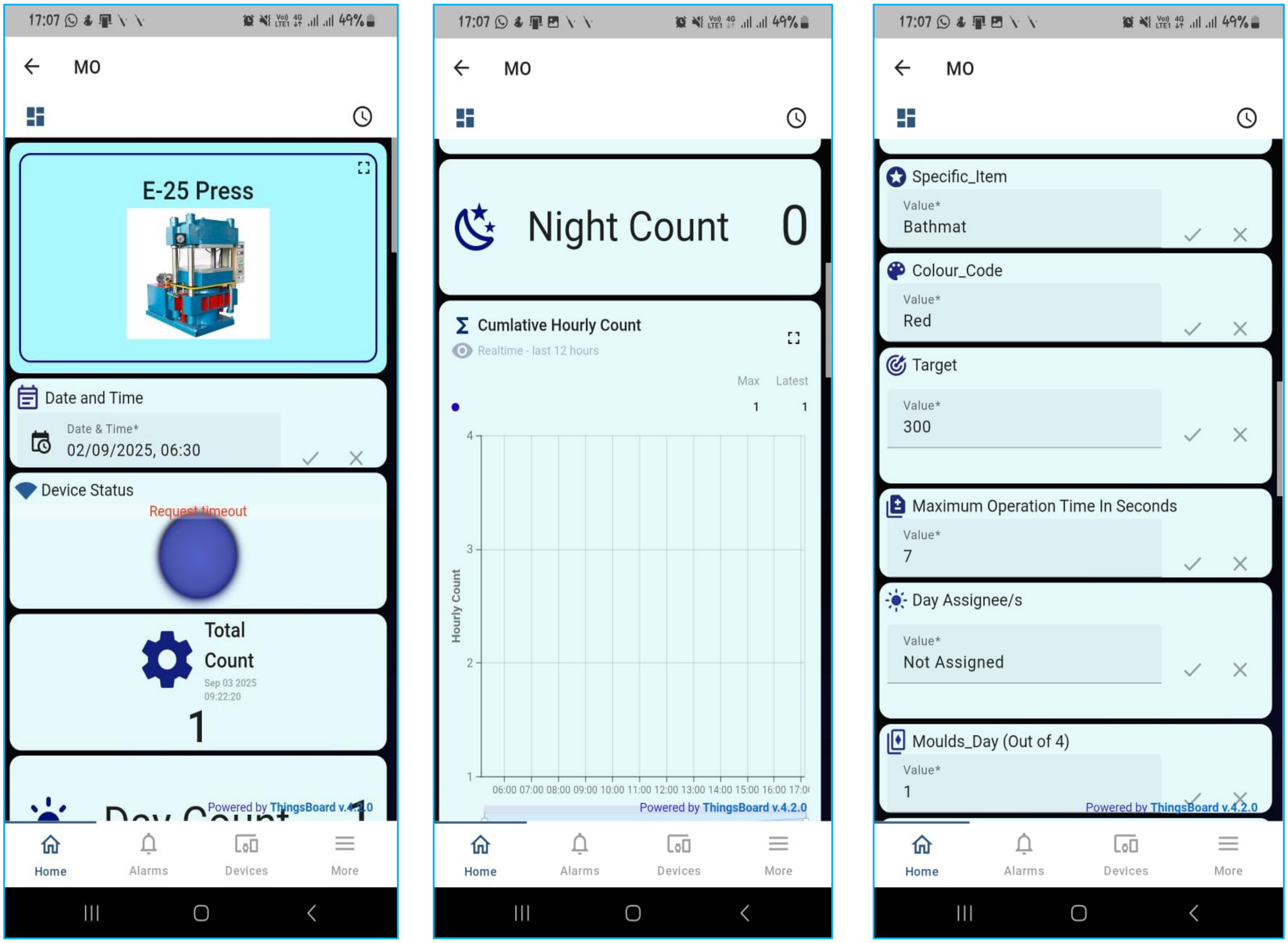


Figure 5: Things-Board Dashboard – Mobile App- I

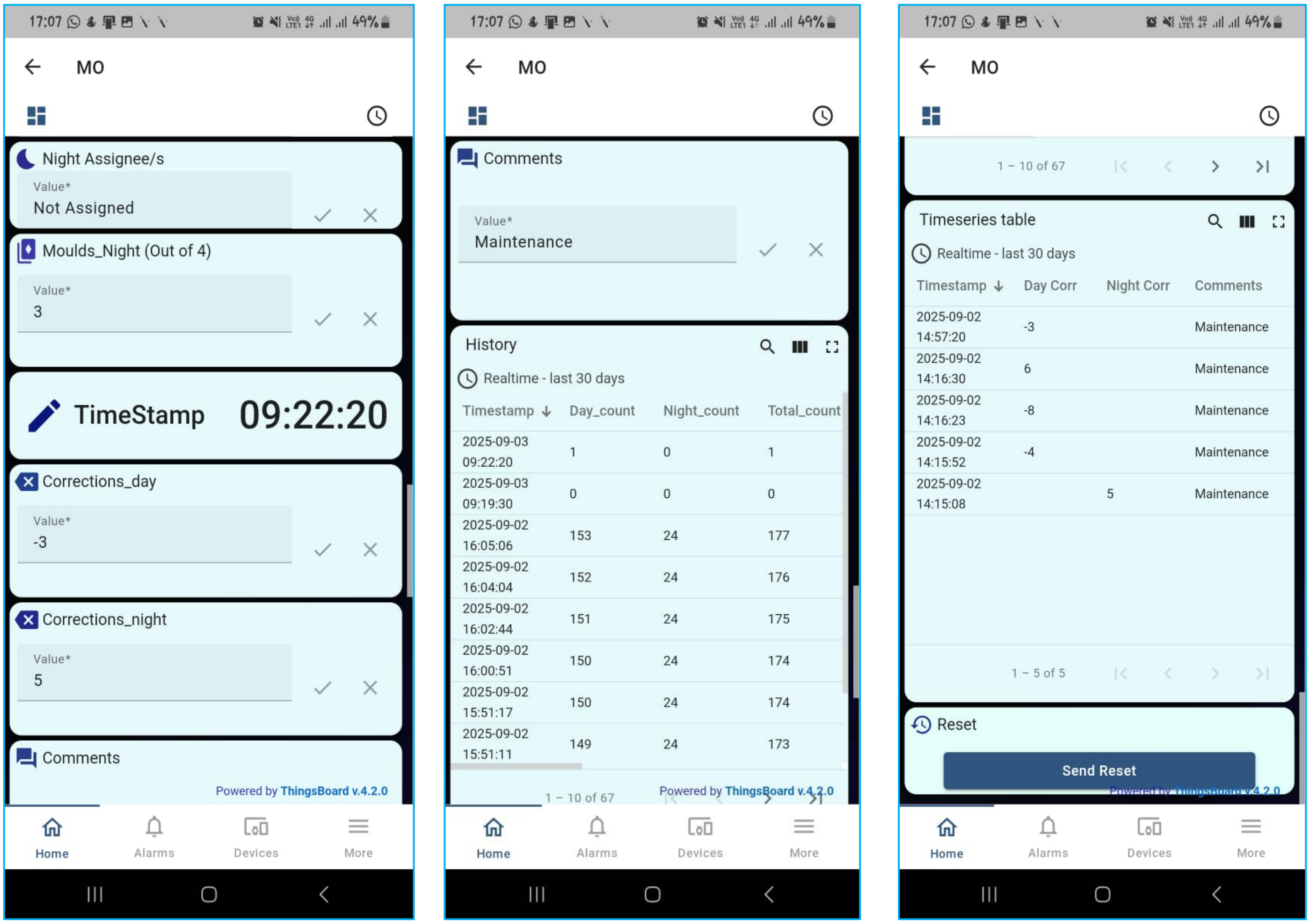


Figure 6: Things-Board Dashboard – Mobile App- II

5 Expected Outcomes

- Real time production hourly tracking.
- Live dashboard updating on web and mobile app.
- Stored data for performance reviews.
- Scalable design for industrial use.

6. Future Enhancements

- Dashboard Improvements with more customized widgets.
- Scale to other machines.
- SMS/Email alerts for production targets.

7. Budget for a one Device:

TABLE 1: BUDGET IN DETAIL WITH THE COMPONENTS

Item	Cost (LKR)
ESP32	1350
AC-DC Power Adapter 230V to 5V	650
1000×700 mm Industrial IP box with hanging	600
Dot Board	50
USB Cable – (Power only)	350
DC Circuit Wires – (20–24 AWG) + Pin Headers (Female)	100
Logic Level Shifter	110
Green Terminal Connectors × 2	30
*Relay Module + Base	500
Total	3740 (With the relay)

*Note: Grey Colour Highlighted: Get from the workshop (Relay with the base)

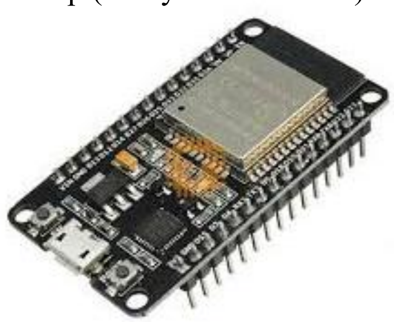


Figure 7: ESP 32 module

Figure 8: AC-DC Power Adapter 230V to 5V with Cabel

Figure 9: Industrial IP box

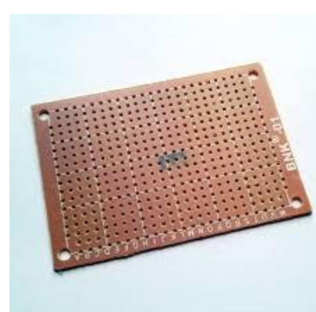


Figure 10: Dot Board

Figure 14: Female Headers

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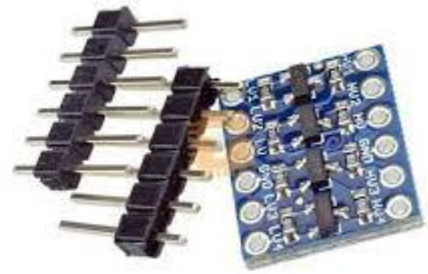

Figure 15: Logic Level Shifter

Figure 17: Green Terminal Connectors

Figure 16: Mechanical Relay with Base

8. Conclusion

This project will provide a low-cost, reliable, and scalable solution to digitize production counting processes and enable real-time monitoring using IoT platform. It can significantly improve data accuracy, visibility, and decision-making in production environments through mobile and the web view.